OPERATING SYSTEMS

INTRODUCTION

BY

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DEFINITION

- An **operating system(OS)** is a software layer that provides a set of services and functionalities to manage computer hardware resources and facilitate the execution of user programs.

- An **operating system(OS)** is system software that manages computer hardware and software resources, and provides common services for computer programs

- It acts as an intermediary between the hardware and software applications
- It provides an interface for software applications to interact with hardware resources efficiently, abstracting away the complexities of underlying hardware details.
- Common Operating systems include Windows, macOS, Linux, Android OS, and Unix.

Main Functions of OS

1. **To allocate resources to processes**
   - **OS** controls how processes may access resources
   - Processes require resources during execution, the **OS** ensures fair and efficient allocation of these resources among competing processes.
   - These resources include CPU time, memory, peripheral devices(I/O devices), network resources and file system resources.
   - It aims is to increase the availability of the computer to its users and maximizing the utilization of the resources.

Main Functions of OS

2. **Provides an Abstraction**
   - The OS simplifies interaction with system resources by hiding hardware complexities and offering a consistent interface for software applications.
   - It creates the illusion of each process having its own CPU and memory space, despite multiple processes sharing the same hardware resources.
   - The OS achieves this abstraction by virtualizing the CPU, allowing processes to operate as if multiple CPUs were available.

Basic Functions of OS

- **Implementing the user interface**: The operating system provides a graphical or command-line interface through which users interact with the computer system.
- **Sharing hardware among users**: The OS manages access to hardware resources such as CPU, memory, and peripherals, ensuring fair allocation among multiple users or processes.
- **Allowing users to share data among themselves**: The OS provides mechanisms for users and processes to exchange data, such as file sharing or inter-process communication.
- **Facilitating I/O operations**: The OS manages input/output devices and provides system calls and drivers for performing I/O operations efficiently.

Basic Functions of OS

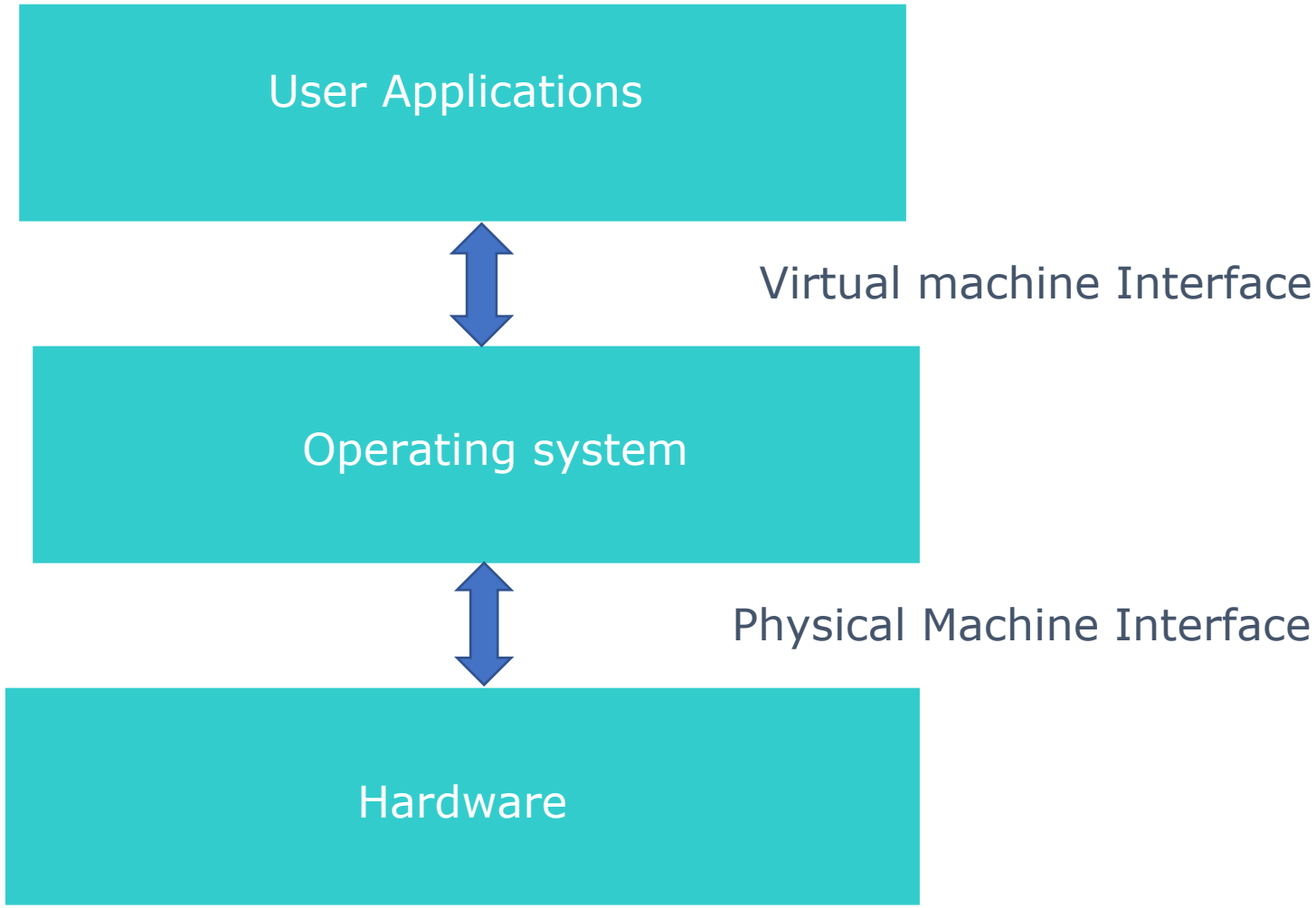
- **Facilitating parallel operations**: The OS supports multitasking and multiprocessing, allowing multiple processes to run concurrently and make efficient use of available hardware resources.
- **Recovering from errors**: The OS implements error-handling mechanisms to detect and recover from faults, crashes, or system failures to maintain system stability and integrity.
- **Organizing data for secure and rapid access:** The OS manages file systems and implements access control mechanisms to organize and protect data, ensuring secure and efficient access.
- **Handling network operations:** The OS provides networking protocols, drivers, and services for communication over networks, enabling connectivity and data exchange between devices.

SYSTEM STRUCTURE

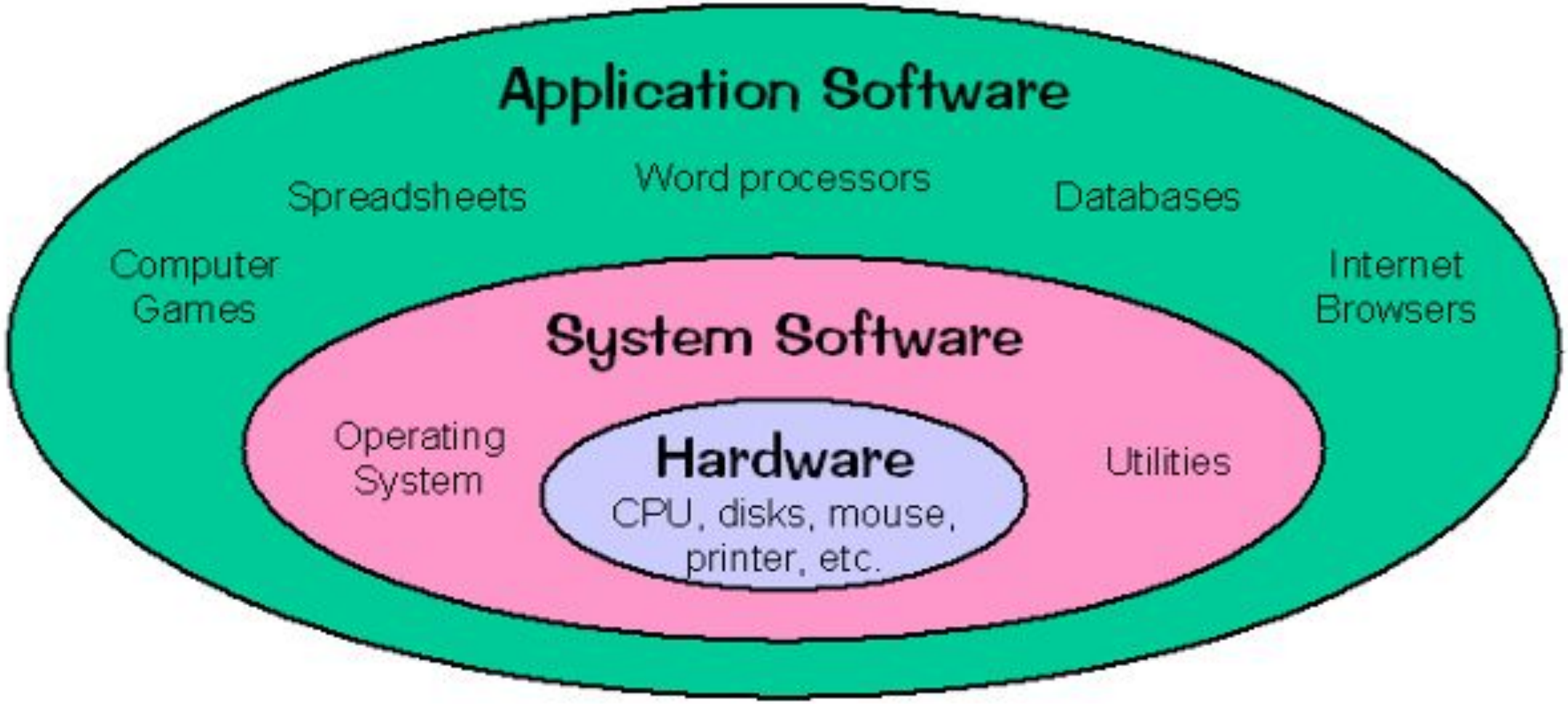
- **SYSTEM** : A collection of hardware, software, and firmware components that work together to perform a specific set of functions or tasks
- **System structure :** Is the organization and arrangement of components within a computer system

  - **SYSTEM SOFTWARE** : Is the foundational software components that provide essential services and functionality to support the operation of a computer system. Such as Operating System, Utilities, Middleware.
  - **OPERATING SOFTWARE :** It specifically refers to the software that manages hardware resources, provides a user interface, and facilitates the execution of application programs
  - **APPLICATION SOFTWARE** : programs and software applications that perform specific tasks or functions for users.

SYSTEM STRUCTURE



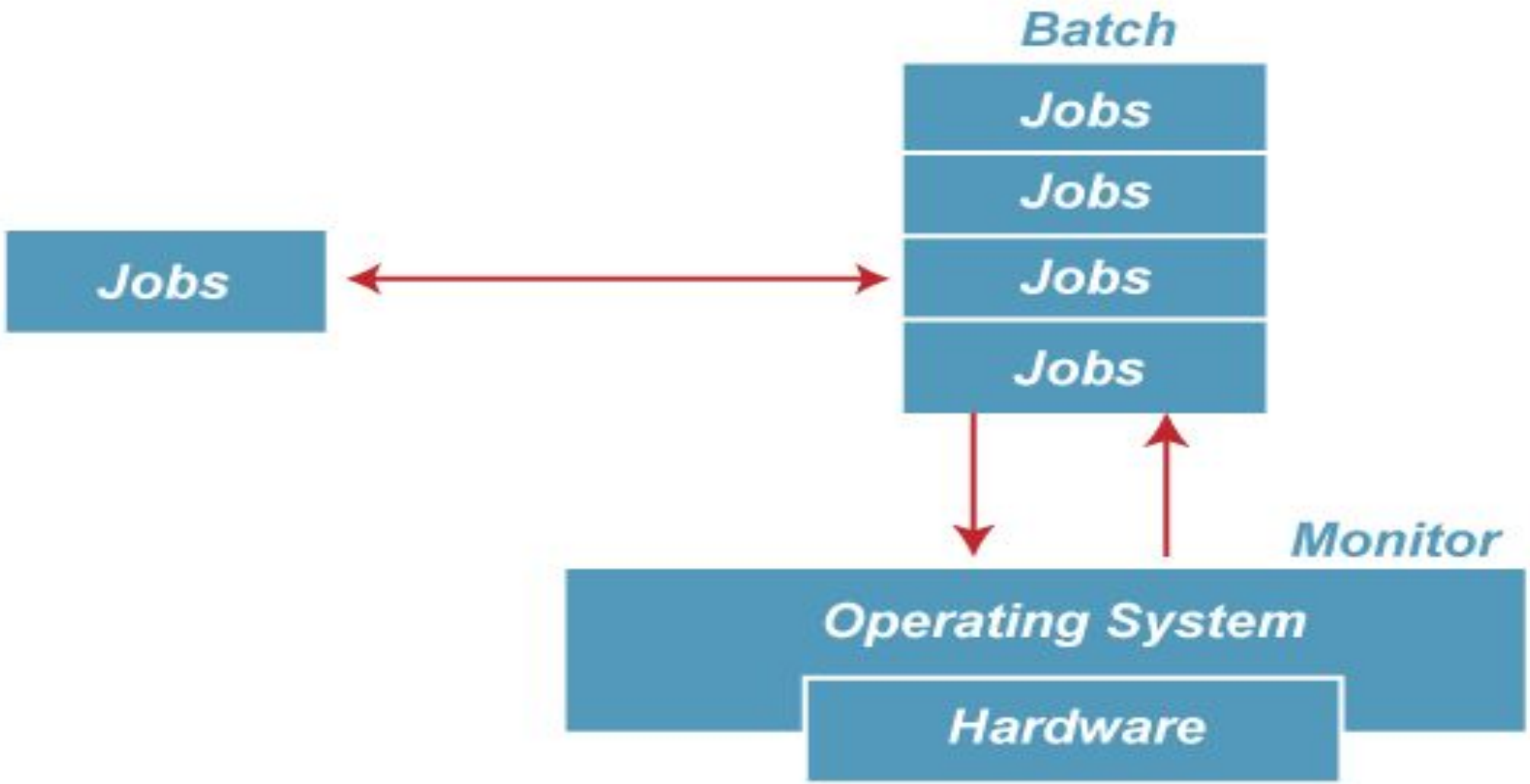
SYSTEM STRUCTURE



Types Of Operating Environments

1. BATCH OPERATING SYSTEM

- It operates in a strict one job-at-a-time manner; within a job, it executes the programs one after another
- Similar types of jobs are batched together and executed in time to further increase economy
- It typically has one main computer or server that receives jobs from different end-user points or terminals
- Once all the jobs in the batch are complete , the Central computer notifies the users of the batch completion
- Once all the jobs in a batch have been executed, the central computer notifies the users or system administrators of the batch's completion.


Batch
Jobs
Jobs
Jobs
Jobs
Jobs
Monitor
Operating System
Hardware

ADVANTAGES

- **It is efficient :** allows for efficient utilization of system resources by executing multiple jobs sequentially
- **Multiple users can share the system: J**obs are typically submitted from different end-user points or terminals
- **Efficient for repetitive tasks :** it automates repetitive tasks and streamline workflows,
- **Efficient resource management :**system ensures optimal allocation of system resources, minimizing **resource contention,** maximizing throughput.

DISADVANTAGES

- **Limited interactivity** : jobs are executed sequentially without user intervention
- **Single Point of Failure**: Failure of the central system can disrupt the execution of all queued jobs

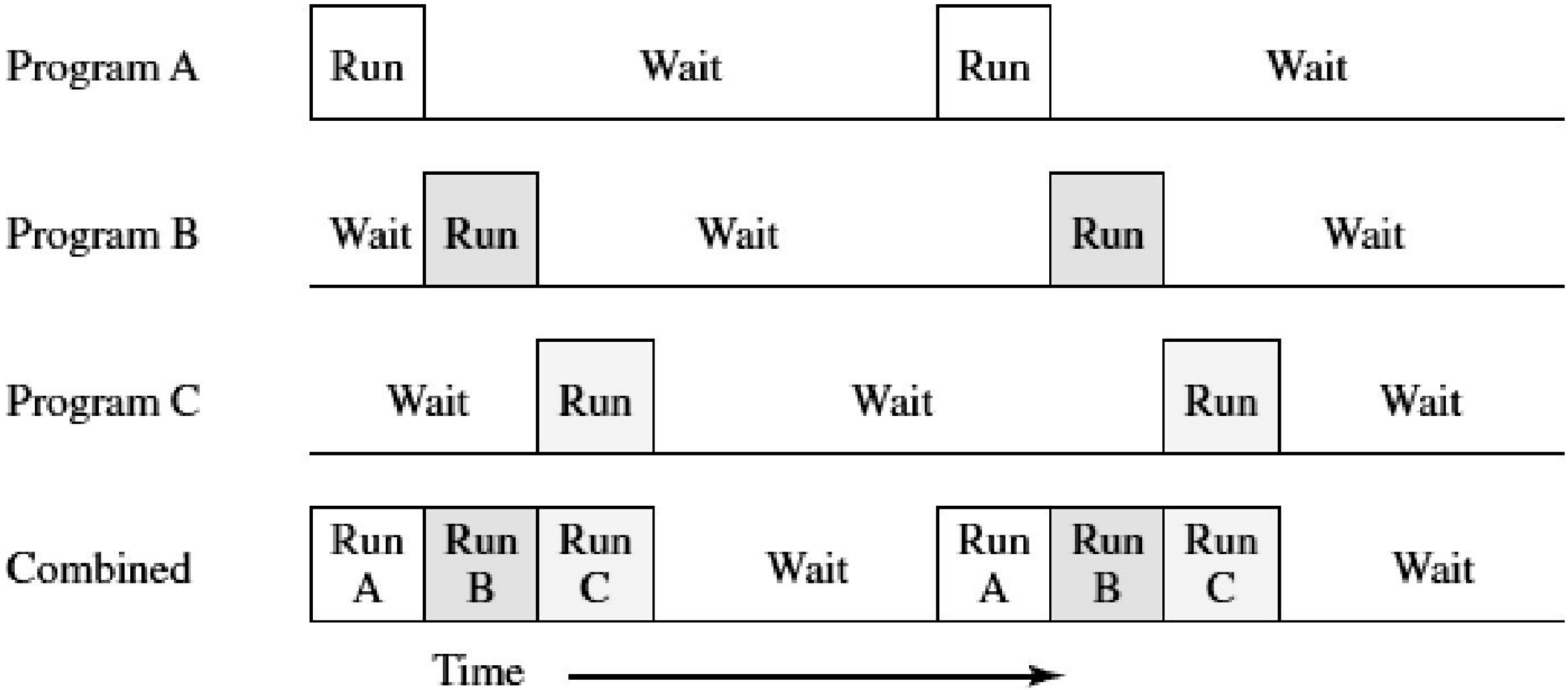
2. **Multiprogramming**

- As machines with more and more memory became available, it became possible to create systems that would load several jobs into memory at once and cycle through them in some order, working on each one for a specified period of time.
- This system is designed to maximize CPU utilization by allowing multiple programs to reside in memory simultaneously and execute concurrently
- Each Process requires CPU time and I/O time, While one program waits for an input/output (I/O) operation to finish, another program utilizes the CPU.

```
-------------------------------------
|            Monitor
|   (more like a operating system)
-------------------------------------
|        User program 1
-------------------------------------
|        User program 2

-------------------------------------
|        User program 3
-------------------------------------
|        User program 4
-------------------------------------
```

- Each of the programs will require CPU time and I/O time. When User program 1 is loaded in the CPU, it is accessing it CPU time and it will leach a point where it requires Some I/O operation.
- When the program reaches this state , the system loads it out of the CPU and into memory until the I/O operations is fulfilled. As this point the next process(User program 2) is loaded into the CPU



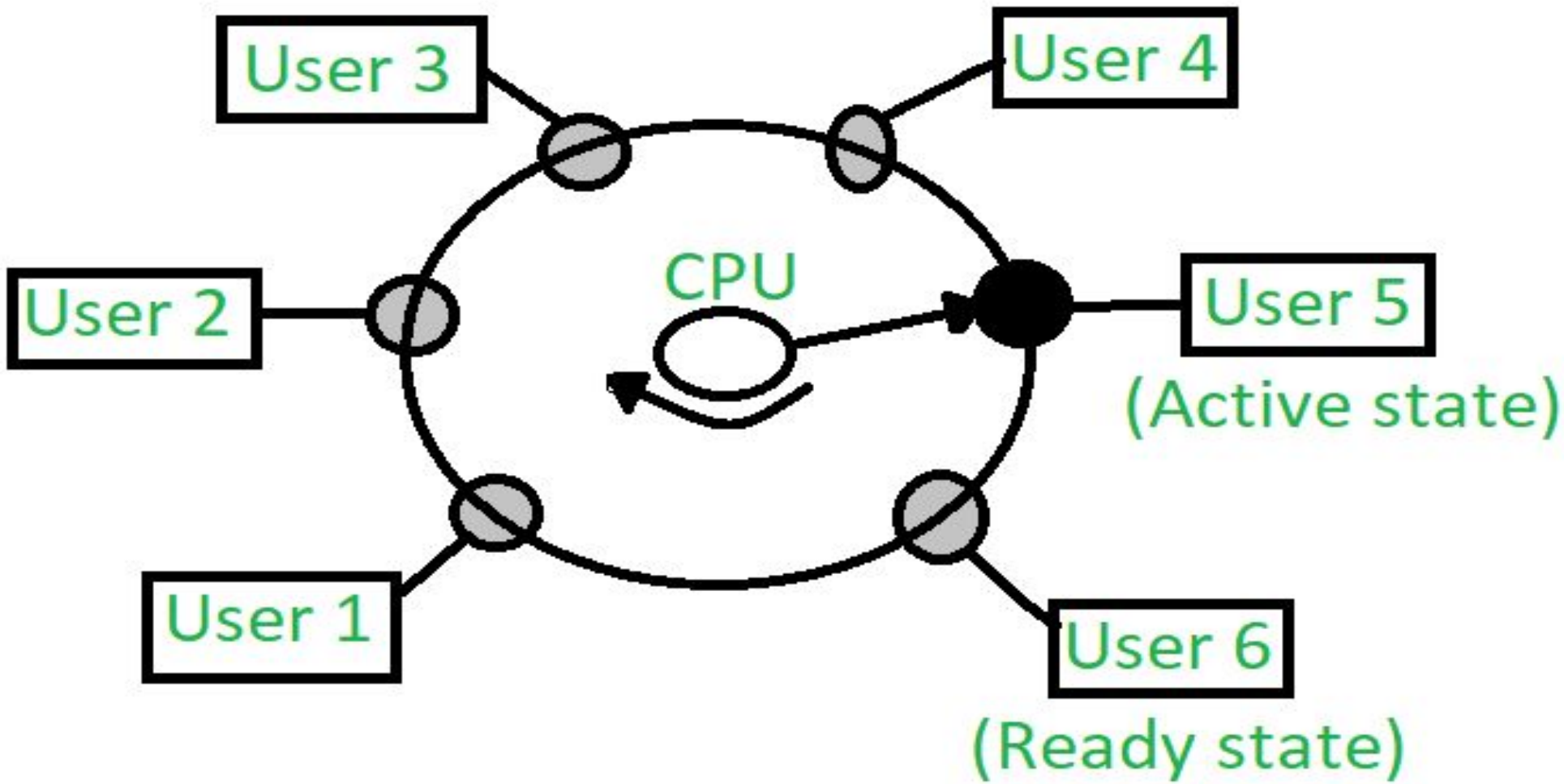
(c) Multiprogramming with three programs

ADVANTAGES

- **Efficient CPU utilization**: there is no point on which the systems CPU is idle
- **Reduced response time:** the CPU is constantly engaged in processing tasks, thereby minimizing waiting times for program execution

3. **Timesharing Systems**

- It is designed to allow multiple users to share a single computer system by dividing the CPU time into small time slices or quanta.
- The CPU switches rapidly between executing different tasks or processes such that the end user can not notice that the CPU is being shared between different users.
- context-switches occurs quickly enough, such that the user has the impression that he or she has direct access to the computer.
- Each user interacts with the system as if they have exclusive access to the computer.
- Interactive processes are given a higher priority


User 3
User 4
User 2
CPU
User 5
(Active state)
User 1
User 6
(Ready state)

ADVANTAGES

- **Resource Utilization**: It allows multiple users to share the same hardware resources (such as CPU, memory, and peripherals) concurrently.
- **Improved Responsiveness**: Time-sharing systems provide quick response times to user requests by rapidly switching between multiple processes.

SINGLE USER SYSTEM

- A system in which only one user can access the computer system at a time.
- Designed to support the needs of a single user, who has exclusive control over the system's resources during their session.
- Single-user operating systems are typically found on personal computers, laptops, and mobile devices
- Examples : Microsoft Windows, macOS

ADVANTAGES

- **Simplicity**: They have fewer features and complexities, making them more straightforward for novice users or those with basic computing needs.
- **Performance**: They are optimized for the specific hardware and software requirements of a single user, resulting in better performance and responsiveness
- **Customization**: Users of single-user systems have the freedom to customize their environment according to their preferences and requirements.

SINGLE TASKING

An operating system that is designed to allow a user to perform only one task at a time.

How it works:

- **Initiation**: The user launches a program by executing its executable file or invoking its command.
- **Loading**: The operating system loads the program into memory from disk, along with any necessary libraries and resources.
- **Execution**: The program is given control of the CPU, and it begins executing its instructions sequentially.
- **Completion/Termination**: Once the program finishes its task or encounters an error, it terminates, and control returns to the operating system.
- **Idle State**: During the execution of the program, the operating system remains idle, waiting for the program to complete before allowing the execution of another program.

MULTI TASKING

- An OS that is designed to allow a single user to perform more than one task at a time.

How it works:

Process Scheduling: The operating system divides CPU time among multiple processes using scheduling algorithms. Each process is allocated a time slice during which it can execute its instructions.

Context Switching: When a time slice for one process expires or it voluntarily relinquishes control (e.g., due to I/O operation), the operating system saves its current state (context), including register values and program counter, and switches to another process.

While only one process runs on the CPU at a time, the rapid switching between processes gives the appearance of parallel execution to users.

Multiprocessor

- A multiprocessor computer is one with more than one CPU.

Categories:

1. **Shared memory multiprocessors**
   - have multiple CPUs, all with access to the same memory. Communication between the processors is easy to implement, but care must be taken so that memory accesses are synchronized.
2. **Distributed memory multiprocessors**
   - multiple CPUs, but each CPU has it's own associated memory.
   - Memory access synchronization is not a problem, but communication between the processors is often slow and complicated.

REAL TIME SYSTEMS

- is an operating system for real-time computing applications that processes data and events that have critically defined time constraints(deadline).
- **Real-time computing applications** are those in which the correctness and timeliness of the system's response are critical for its operation
- A **real-time operating system** is used to implement a computer application for controlling or tracking of real-world activities.

NETWORKED VS DISTRIBUTED

- **Networked systems**
  - consist of multiple computers that are networked together, usually with a common operating system and shared resources.
  - Users, however, are aware of the different computers that make up the system.

- **Distributed systems**
  - Consists of multiple computers but differ from networked systems in that the multiple computers are transparent to the user.
  - Often there are redundant resources and a sharing of the workload among the different computers, but this is all transparent to the user.

HISTORY/EVOLUTION OF OPERATING SYSTEMS

- OS will evolve overtime as:
  1. **Hardware upgrades** : Introductions of new computing chips , improvement of the current computing chips will require OS to also be up to standards
  2. **New Service Demand**: The OS evolves to cuter for these new demands
  3. **Fix errors:** There is a requirement to fix these errors

First Generation (1940s-1955s):

- **No Operating Systems:** Initially, computers were programmed using machine language or assembly language without the concept of an operating system.
- **Single User, Single Program:** Early computers were dedicated to running a single program at a time, with users manually loading programs and data. Each program contained Operating instructions.
- **Use of Vacuum Tubes**: The foundational electronic components powering the computational capabilities of these early machines

Second Generation (1955s-1965s):

- **Simple Batch Systems:** introduction of simple batch processing capabilities, managing job scheduling, and input/output operations.
- **Limited User Interaction:** User interaction was minimal, primarily through punched cards.
- **Use of Transistors**: vacuum tubes were gradually replaced by transistors, which were smaller, more reliable, and consumed less power, leading to improvements in computer size, speed, and efficiency.

Third Generation (1965-1980)

- **Time-Sharing Systems:** Time-sharing operating systems allowed multiple users to interact with the system simultaneously.
- **Multiprogramming:** multiple programs resided in memory concurrently, enabling efficient resource utilization.
- **Interactive Computing:** Introduction of interactive computing environments.
- **Development of High-Level Languages:** Proliferation of high-level programming languages like FORTRAN, COBOL, and ALGOL.

Fourth Generation (1980-Present)

- **Personal Computer :** The birth of Microsoft and windows operating systems played a major role in the development of personal computers.
- **GUI-Based Systems:** GUI-Based Systems introduced visual interfaces with icons and windows, making it easier for users to interact with computers through simple point-and-click actions.
- **Networking:** Networking enabled computers to connect and share resources, facilitating collaboration and communication over local and wide area networks, revolutionizing how information is accessed and shared.

Fourth Generation (1980-Present)

- **Mobile Operating Systems:** Rise of mobile operating systems such as iOS, Android, and Windows Mobile for smartphones and tablets.
- **Virtualization and Cloud Computing:** Virtualization technologies and cloud computing platforms revolutionized how computing resources are provisioned, allowing for the creation of virtual instances and the delivery of scalable services over the internet
- **Use of Integrated circuits:** use of integrated circuits revolutionized computing by enabling the miniaturization of components, leading to smaller, more powerful, and energy-efficient computing devices